

Equipment Insights — Scenario Report

Date: 6 July 2026 (Overview); **8 July 2026** (Trends & Incidents addendum)

Scope:

- **Part A — Overview:** 10 devices across hydraulic, traction, escalator, and moving walkway
- **Part B — Trends:** 5 devices × calendar presets (`last_week` ... `last_2_months`)
- **Part C — Incidents:** 5 devices × calendar presets on the incident log tab

Sources: Staging dashboard insight cards + Jobs DB (`device_insight_cache` , `device_period_stats` ; see `sql-queries_llm.txt`)

UI captures: `llm-insights.txt` (Overview 6 Jul; Trends/Incidents 8 Jul 2026)

Executive summary

Facilities managers oversee hundreds of elevators, escalators, and moving walkways. Each unit generates **daily trip counts, door cycles, motor starts, runtime, alert logs, and peer comparisons** — data that is accurate but spread across charts, tables, and incident history. Most managers do not have time to inspect every metric on every device every week.

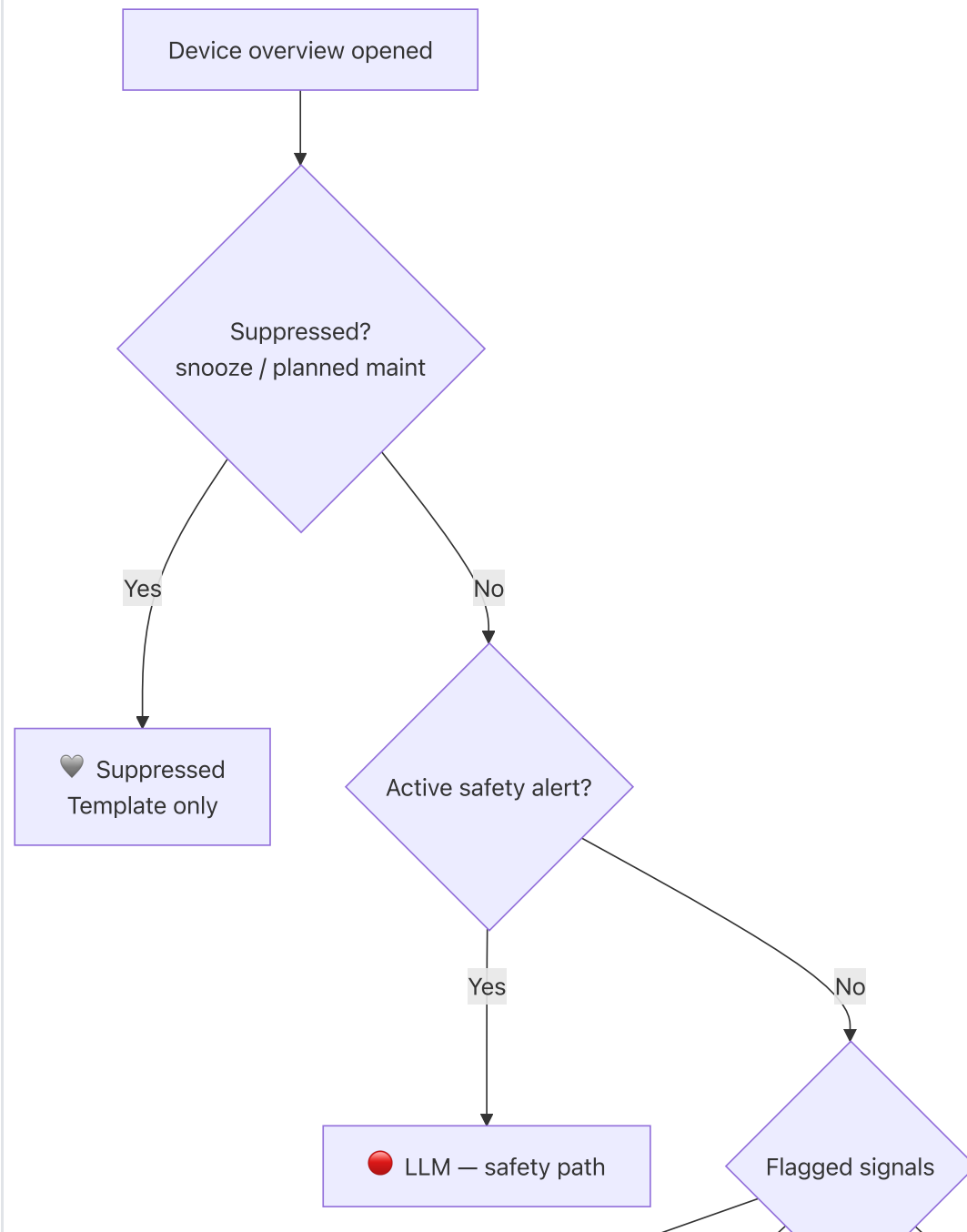
Equipment Insights solves this by surfacing a single **insight card** on each unit's overview page. The card answers three questions at a glance:

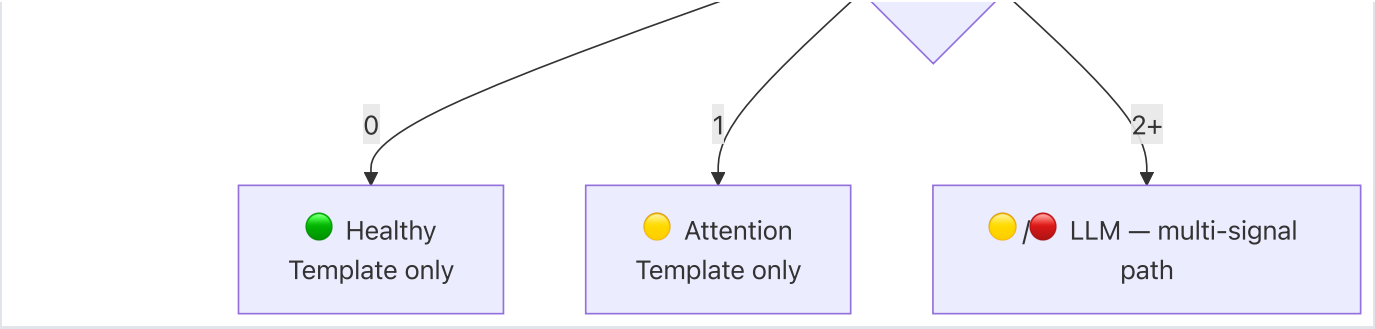
1. **Is this unit okay right now?** (Healthy / Attention / Degraded / Suppressed)
2. **What changed?** (headline + up to 3 signals with numbers)

3. **What should I keep an eye on?** (watch items — observational, not prescriptive)

The system uses a **gate** to decide whether a fixed template is enough or whether an LLM should write the card:







Key finding from this study: Templates work well for **healthy, single-trend**, and **suppressed** units. The LLM adds clear customer value when **multiple metrics move in conflicting directions** or when **alerts and usage data tell different parts of the same story** — patterns a manager would only catch by cross-referencing several dashboards manually.

v1 tab extension (8 Jul): Trends and Incidents reuse the same card pattern but scope answers to the **selected calendar period** and the tab context (charts vs. incident table). Rich **healthy templates** now cite peer standing, period deltas, and sub-bucket trends (Trends). **Attention templates** carry multiple quantified bullets without LLM when only one metric flags. Incidents LLM paths digest **volume, downtime, top fault, and clustering** — the patterns a manager would otherwise infer from scrolling outage rows.

What the customer sees vs. what they'd have to do without it

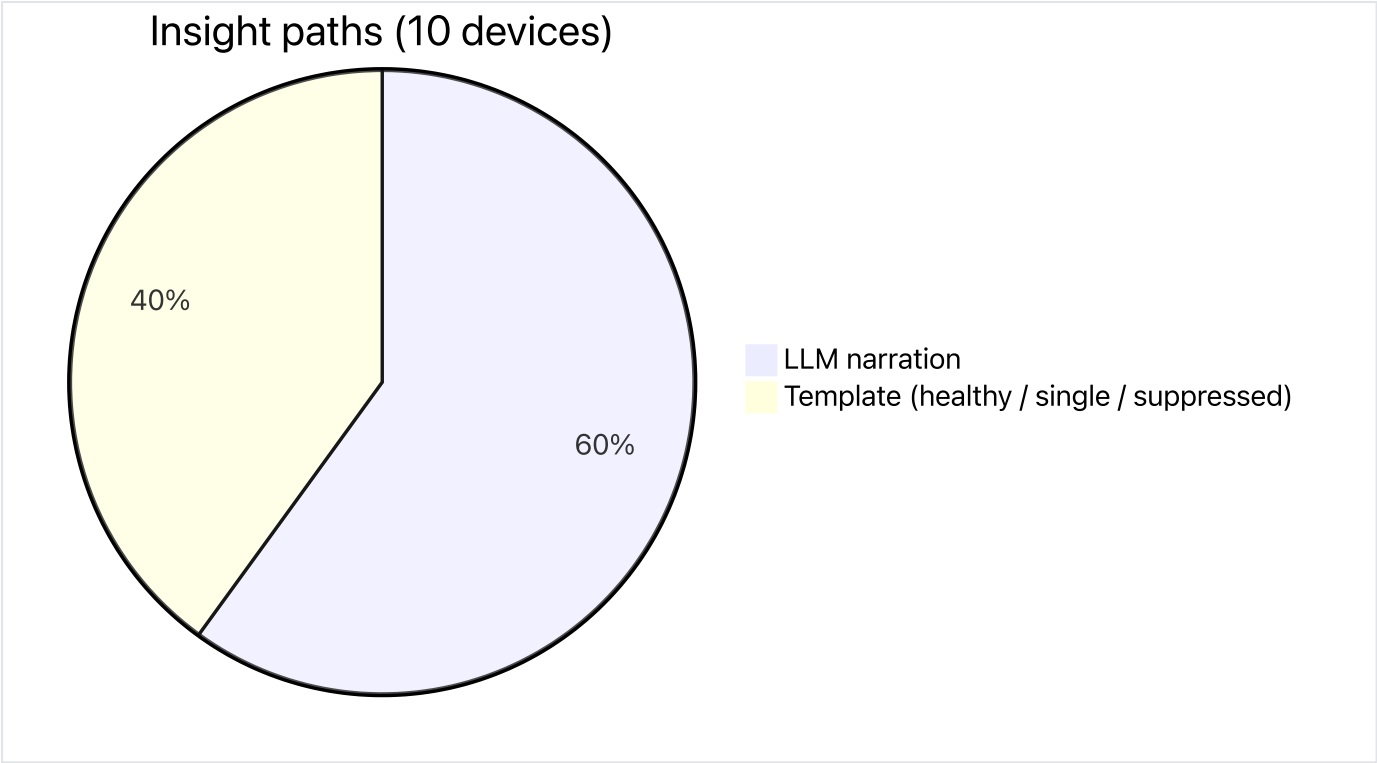
Without Equipment Insights	With Equipment Insights
Open equipment page, scan 11 trend charts	Read one card (~15 seconds)
Check alert history for recurring faults	Card states alert frequency + recovery time
Mentally compare this week vs. last month	Card cites baseline vs. current with % change

Cross-reference trips vs. door cycles vs. runtime	Card highlights diverging trends in one sentence
Wonder if a quiet unit is healthy or offline	Card flags coordinated usage drops across metrics
Investigate units under planned maintenance	Card shows “under planned maintenance” — no false alarms

Scenario coverage matrix

Ten devices were selected to maximize **equipment type** and **insight path** diversity.

#	Unit	Type	Building	UI state	Path	LLM?
1	A06	Hydraulic	EWR Terminal A	⦿ Attention	Single metric template	No
2	Elevator #15	Traction	Woolworth Building	● Healthy	Healthy template	No
3	A03	Hydraulic	EWR Terminal A	⦿ Attention	Multi-signal LLM	Yes
4	A04	Hydraulic	EWR Terminal A	⦿ Attention	Multi-signal LLM	Yes
5	PE-A #2	Traction	JFKIA PA Bldg. 14	⦿ Attention	Multi-signal LLM	Yes
6	A01	Hydraulic	EWR Terminal A	⦿ Attention	Multi-signal LLM	Yes
7	ES - HB NO. 1- UP	Escalator	JFK AirTrain Howard Beach	● Degraded	Multi-signal LLM + alerts	Yes
8	BW-21	Moving walkway	Concourse B	💔 Suppressed	Suppression template	No
9	Elevator D	Hydraulic	World Trade Center	● Degraded	Multi-signal LLM	Yes
10	CB-MW02	Moving walkway	LGA Terminal B	⦿ Attention	Single metric template	No



Health state legend

Badge	Meaning for the customer
● Healthy	No flagged metrics or alerts. No action implied.
● Attention	Something shifted vs. this unit's own baseline or peers. Worth a look, not necessarily an emergency.
● Degraded	Stronger signal — recurring alerts, extreme shifts, or safety-tier activity. Prioritize review.
♥ Suppressed	Unit is snoozed or under planned maintenance. Insights intentionally muted.

Case studies — Part A (Overview tab)

Each section follows the same structure:

- **The scenario** — what was happening in the data
 - **Without the insight card** — what the manager would need to discover manually
 - **What the card showed** — exact UI text from 6 Jul 2026
 - **How the insight helps** — why this saves time and surfaces the right story
-

1. A06 — Hydraulic elevator, EWR Terminal A

Device ID: EC4F9F1B-1CE4-42D9-80B2-3796B0C69174

The scenario

Average trip time has been **rising for 2 consecutive weeks** — a single metric trend crossing the system's threshold. No alerts fired. One flagged signal → template path (no LLM).

*Note: This unit showed **Healthy** in the database a week earlier. Opening the dashboard on 6 Jul refreshed the insight — a real example of live monitoring.*

Without the insight card

The manager would need to open the **Average trip time** chart, notice the upward slope over ~14 days, and decide whether a ~2-week rise is meaningful vs. normal variance.

What the card showed

Equipment Insight

● Attention

Average trip time has been rising for 2 weeks.

- Average trip time has been rising for 2 weeks.

Updated 6 Jul, 11:40

How the insight helps

A **single plain-English line** replaces chart inspection. The manager knows *which metric* and *how long* without navigating trend views. For one signal, a template is sufficient — no LLM cost needed.

2. Elevator #15 — Traction elevator, Woolworth Building

Device ID: E292092B-2618-40B2-9300-599EBF34C82D

The scenario

No metrics or alerts crossed flag thresholds. Zero signals → healthy template.

Without the insight card

The manager sees a unit with data but no obvious problems. They might still wonder: “Is anything trending?” — and would need to scan multiple charts to confirm all-clear.

What the card showed



How the insight helps

Explicit confirmation of normal operation. In a fleet of hundreds of units, a green “Operating normally” card is valuable negative information — the manager can move on without drilling in.

3. A03 — Hydraulic elevator, EWR Terminal A

Device ID: 475A4B8B-4468-4793-B40D-1E8C31FC1510

The scenario

Three related metrics flagged over 2–3 weeks:

Metric	Direction	Magnitude
Trips per day	↑ Rising	+3.9% above baseline, 3 weeks
Motor starts	↑ Rising	+2.9% above baseline, 2 weeks
Runtime	↑ Rising	+3.3% above baseline, 2 weeks

Door cycle-to-trip ratio also elevated at **1.196**. Multiple signals → **LLM path**.

*A week earlier, only one metric (door open time) was flagged and a template was used.
Fresher data on 6 Jul crossed the multi-signal threshold — the card upgraded to full LLM
narration.*

Without the insight card

The manager would need to:

1. Notice rising trips on the trips chart
2. Separately notice rising motor starts
3. Separately notice rising runtime
4. Connect them as **coordinated increased utilization**
5. Check whether door cycle ratio is also elevated

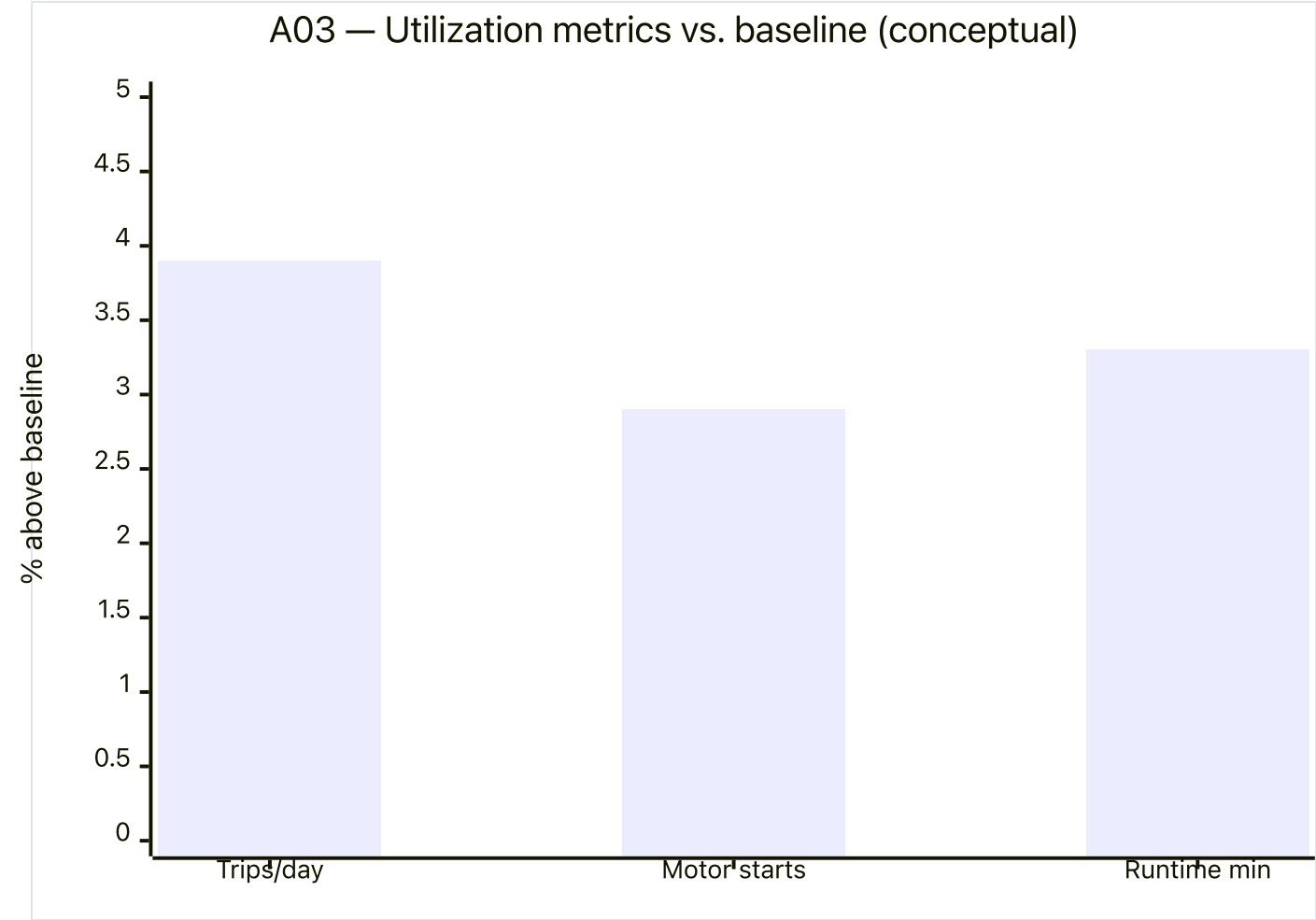
That is 4+ chart interactions and mental synthesis.

What the card showed

Equipment Insight	🟡 Attention
Rising trip frequency and motor starts over two to three weeks	
• Trips per day increased 3.9% above self-baseline and have risen for 3 consecutive weeks, currently at 1,474.6 trips per day. • Motor starts rose 2.9% above self-baseline over the past 2 weeks, now at 6,138 starts. • Runtime increased 3.3% above self-baseline over 2 weeks to 3,230.6 minutes.	
Watch: Trip frequency and motor start rate sustained elevation—confirm if demand increase is temporary or sustained.	
Watch: Door cycle trip ratio remains elevated at 1.196 despite rising overall activity.	
Updated 6 Jul, 11:44	

How the insight helps

The LLM **bundles correlated utilization increases** into one headline and cites exact numbers. Watch items nudge the manager to distinguish **temporary demand spike** vs. **sustained load** — without saying “schedule maintenance.” The door-cycle ratio watch item flags a secondary pattern the manager might miss when focused only on trip volume.



4. A04 — Hydraulic elevator, EWR Terminal A

Device ID: 9191D75D-5994-4414-999B-2F6B5A327A72

The scenario

A **positive operational story** with a subtle twist:

Metric	Change over 3 weeks
Trips per day	↑ +7.3% (1,151 → 1,235)
Door cycles	↑ +6.2% (11,489 → 12,206)
Average trip time	↓ −0.9% (20.6s → 20.4s)

More volume **and** slightly faster trips — three metrics, LLM path.

Without the insight card

Rising trips and cycles look like good news in isolation. The slight trip-time **decrease** is easy to overlook on a busy dashboard. The manager might not connect “higher throughput + stable speed” as a coherent pattern.

What the card showed

Equipment Insight	Attention
Trips and door cycles rising over 3 weeks; trip time falling	
• Trips per day increased to 1235.6 from baseline 1151.6 over 3 weeks, a 7.3% rise. • Door cycles increased to 12206 from baseline 11489.5 over 3 weeks, a 6.2% rise. • Average trip time decreased to 20.4 seconds from baseline 20.6 seconds, a 0.9% decline over 3 weeks.	
Watch: Sustained increase in operational frequency	
Watch: Concurrent decrease in trip time while handling higher volume	
Updated 6 Jul, 11:44	

How the insight helps

The headline captures a **counter-intuitive pairing**: more trips *and* faster trips. That is the kind of cross-metric read that normally requires an analyst. Watch items frame it observationally — useful for capacity planning without alarmism.

5. PE-A #2 — Traction elevator, JFKIA Port Authority Bldg. 14

Device ID: 2509AA46-DB11-45C5-9501-598A6C70D08D

The scenario

A **coordinated usage decline** across six metrics over 3 weeks (~15–18% drops):

Metric	Baseline → Current	Change
Trips/day	421 → 347	−17.5%
Total distance	9,335 → 7,592 ft	−18.6%
Door cycles	2,964 → 2,494	−15.9%
Runtime	498 → 414 min	−17.0%
Motor starts	2,921 → 2,432	−16.7%
Door cycle/trip ratio	—	+0.7% (slight rise)

A week earlier, this unit was **Degraded** with an additional `door_open` alert (4 consecutive periods). By 6 Jul the card showed **Attention** with metrics only — severity shifted as underlying signals changed.

Without the insight card

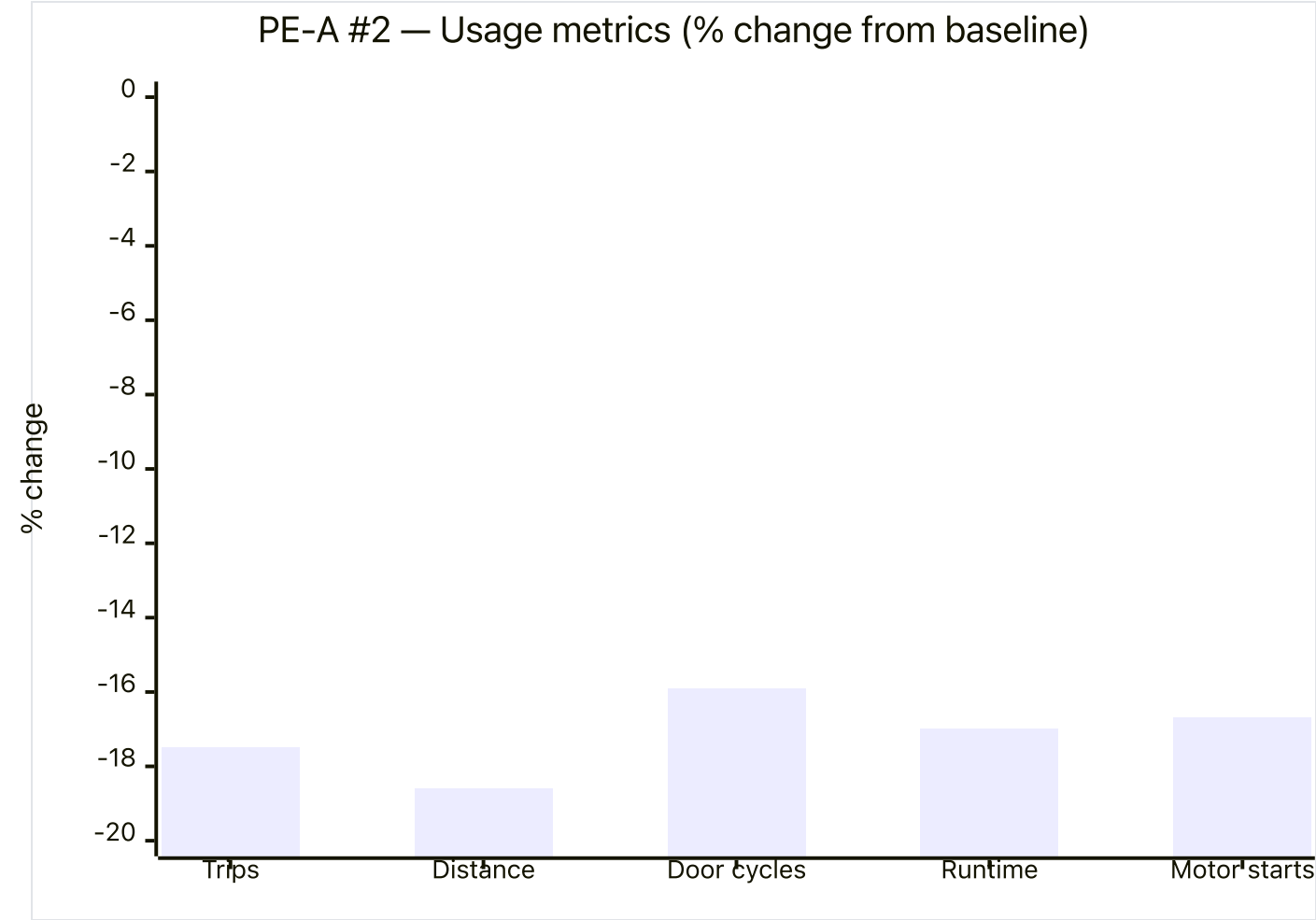
Each metric chart alone shows a modest decline. The manager might notice trips are down but miss that **distance, cycles, runtime, and motor starts all fell in parallel** — the signature of reduced utilization or partial outage, not a single noisy metric.

What the card showed

Equipment Insight	🟡 Attention
PE-A #2 usage falling across all metrics over 3 weeks	
• Trips per day down 17.5% (347 vs 421); total distance down 18.6% (7,592 vs 9,335). • Door cycles down 15.9%; door cycle-to-trip ratio rose 0.7% – fewer but slightly less efficient ops. • Runtime down 17.0%; motor starts down 16.7%.	
Watch: Sustained decline across six independent metrics suggests reduced demand or operational constraint	
Watch: Door cycle efficiency ratio uptick despite lower overall activity warrants monitoring	
Updated 6 Jul, 11:45	

How the insight helps

This is the **strongest “don’t make me dig” example** for traction units. One headline — “usage falling across all metrics” — replaces checking six charts. The watch item names the two hypotheses a manager would form anyway (demand vs. constraint) without prescribing action.



6. A01 — Hydraulic elevator, EWR Terminal A

Device ID: 3ADEB45E-9BD7-44D6-8FC6-14FEE641CC7E

The scenario

Diverging door behavior — the hardest pattern for a manager to spot quickly:

Metric	Direction	Detail
Trips per day	↓ Falling	1,946 → 1,736
Door cycle-to-trip ratio	↑ Rising	0.99 → 1.19
Average door cycle time	↑ Rising	10.05s → 10.85s
Runtime	↓ Falling	4,700 → 4,133 min (−12%)

Fewer trips, but doors cycling **more per trip** and taking **longer** — five flagged metrics, LLM path.

Without the insight card

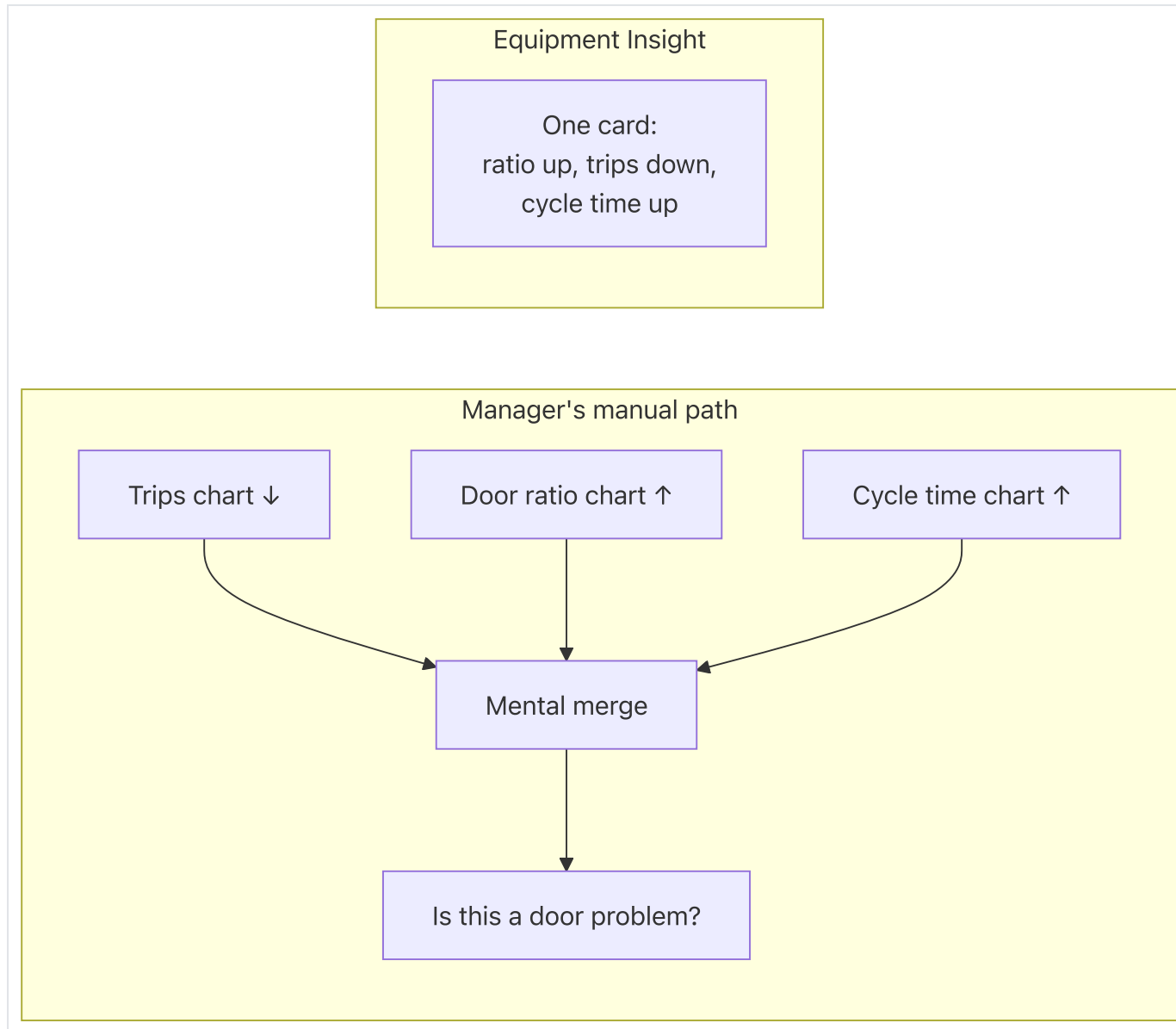
Trips-down looks like low demand. Door-ratio-up is on a different chart. Cycle-time-up is on a third. Connecting “fewer trips + worse door efficiency” requires domain expertise **and** time — exactly what facilities managers lack during a busy shift.

What the card showed

Equipment Insight	🟡 Attention
Door cycle time rising while trip frequency falls over 3 weeks	
• Door cycle-to-trip ratio increased to 1.19 from 0.99 while trips fell to 1,736 from 1,946 – doors cycling more relative to actual trips. • Average door cycle time rose to 10.85s from 10.05s. • Runtime decreased to 4,133 from 4,700 (–12%).	
Watch: Door cycle timing degradation paired with lower trip volume suggests mechanical friction or control logic drift	
Watch: Sustained 3-week trend in door cycle-to-trip ratio warrants continued observation	
Updated 6 Jul, 11:47	

How the insight helps

The LLM's core value here is **synthesis of opposing trends**. The headline tells the story a chart-by-chart review would take 10+ minutes to assemble. For hydraulic elevators — where door mechanics dominate reliability — surfacing door-ratio drift alongside falling trips is precisely the insight a customer pays for.



7. ES - HB NO. 1- UP — Escalator, JFK AirTrain Howard Beach

Device ID: 4EE9BD2E-B19A-44C9-A5E2-C95197AB96E2

The scenario

Two signal types combined — **alert recurrence** plus **metric anomaly**:

Signal	Detail
Escalator motor alert	2 occurrences this period; present 3 consecutive periods ; avg recovery 134.5 minutes
Total distance	Fell to 0.0 from baseline ~794,726 (sensor or operational issue)

Alert + metric → LLM path. UI badge: **Degraded**.

Without the insight card

The manager would check the alerts tab for motor faults, then separately notice distance telemetry flatlined, then mentally assess whether the unit is actually running. Three different views, no automatic link.

What the card showed

Equipment Insight	Degraded
Escalator motor alerts recurring; distance tracking dropped to zero.	
- Escalator motor alert occurred 2 times this period, present 3 consecutive periods, avg recovery 134.5 minutes. - Total distance fell to 0.0 over 3 weeks, down from baseline of 794,726 miles.	
Watch: Motor alert recovery time and frequency over next reporting period	
Watch: Distance metric sensor or data collection status	
Updated 6 Jul, 11:53	

How the insight helps

The **red Degraded badge** prioritizes this unit in a fleet scan. The card links **recurring motor alerts** with **dead distance telemetry** — prompting the manager to ask “is this unit running, or is the sensor broken?” without reading raw alert JSON or distance logs. This is the clearest escalator case in the study.

8. BW-21 — Moving walkway, Concourse B

Device ID: 5B9A752F-FB88-4A60-B4E8-C44E3A78D255

The scenario

Unit has an **active planned maintenance snooze** (since 1 Jul 2026; note: “Down for power bump issue”). Suppression active → template only, no LLM, no false anomaly flags.

Without the insight card

A suppressed unit might still show stale alerts or odd metrics in other tabs. The manager could waste time investigating a unit that is **intentionally offline**.

What the card showed

Equipment Insight

Suppressed

This unit is currently under planned maintenance.

Updated 6 Jul, 11:54

How the insight helps

Trust through silence. The customer knows the system respects their maintenance workflow. No LLM narration, no misleading “attention” flags on a unit they deliberately took offline. Critical for moving walkways and escalators with long planned outages.

9. Elevator D — Hydraulic elevator, World Trade Center

Device ID: 1CECA405-0181-4571-9488-1214D8F3A500

The scenario

Severe operational collapse over 2 weeks:

Metric	Baseline → Current	Change
Motor starts	1,707 → 268	−84%
Total distance	29,293 → 4,769 ft	−84%
Door cycles	2,298 → 393	−83%
Trips per day	379 → 430	+13% (inverse!)

The last row is the subtlety: trips *count* rose slightly while every usage proxy collapsed — suggesting longer occupancy periods or a unit barely moving. UI: **Degraded**.

This device was selected as a known safety fixture. On 6 Jul the card reflected metric collapse rather than an active safety alert — the underlying condition may have changed since initial data discovery.

Without the insight card

An 84% motor-start drop is visible on one chart. The trips-per-day **increase** on another chart looks contradictory. Without synthesis, a manager might misread the unit as “busy” when it is largely idle.

What the card showed

Equipment Insight	Degraded
Elevator D usage and operational metrics declining sharply over two weeks	
• Motor starts fell 84% (1707 → 268). • Total distance dropped 84% (29293 → 4769 ft). • Door cycles declined 83% (2298 → 393), while trips per day rose 13% (379 → 430) – fewer but longer occupancy periods.	
Watch: Verify operational status – intentional maintenance, service suspension, or demand?	
Watch: Monitor for resumption of normal baselines once service status clarified	
Updated 6 Jul, 11:56	

How the insight helps

The LLM resolves an **apparent contradiction** (trips up, everything else down) into a plain hypothesis: fewer but longer occupancy periods. Watch items steer toward **status verification** — the right first step when a World Trade Center elevator shows 84% usage collapse. A manager gets this in one read instead of a forensic session across metrics.

10. CB-MW02 — Moving walkway, LGA Terminal B

Device ID: 908F0F62-503F-415A-AFA7-B32AE8694E8D

The scenario

Motor starts rising for **2 consecutive weeks** — single metric trend. One signal → template path. (This unit also had 62 `escalator_motor` alerts in the prior 90 days, but none crossed the current-period multi-signal gate.)

Without the insight card

The manager would open motor-start trends, confirm the 2-week rise, and decide if it matters for a moving walkway at LGA Terminal B.

What the card showed

Equipment Insight

● Attention

Motor starts has been rising for 2 weeks.

- Motor starts has been rising for 2 weeks.

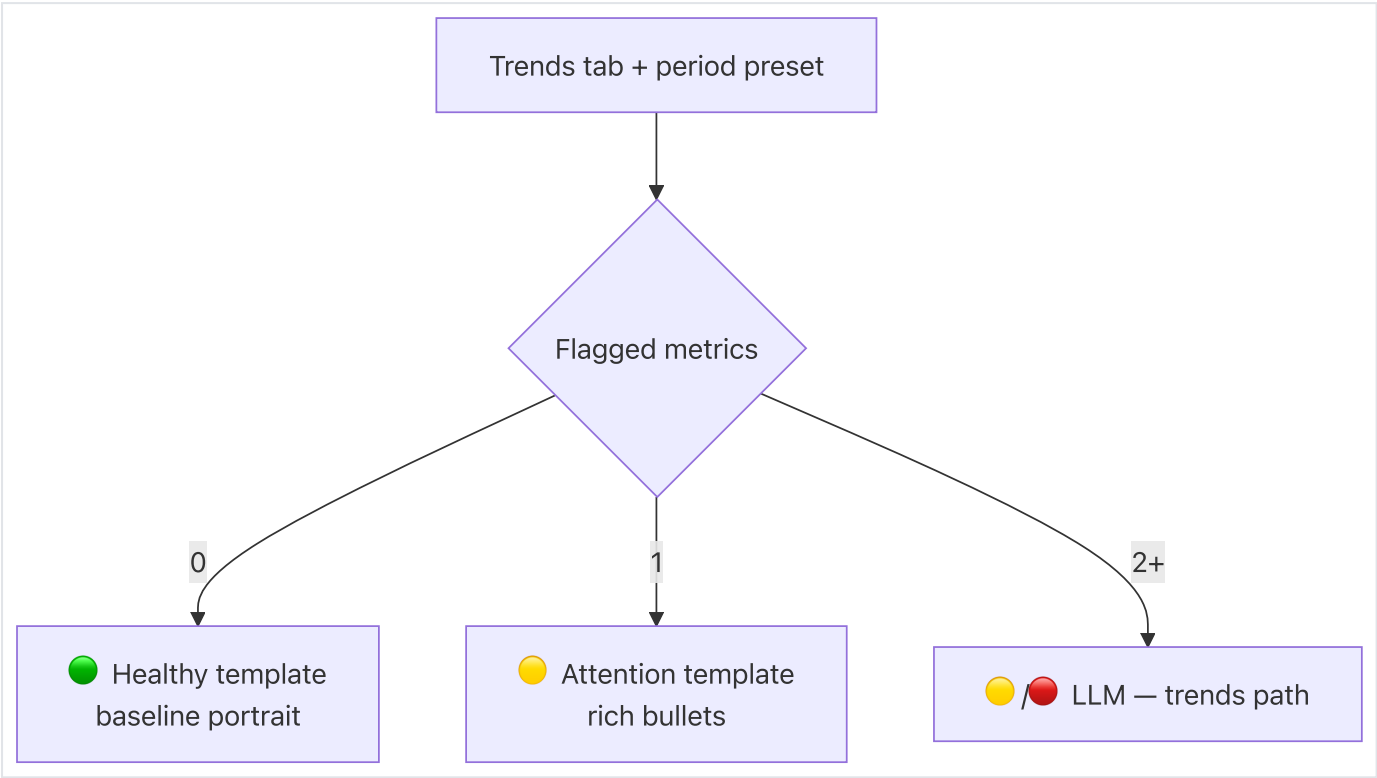
Updated 6 Jul, 11:57

How the insight helps

Same pattern as A06: a **quick template nudge** for moving walkways. Enough to prompt a glance at motor health without LLM overhead. Pairs well with BW-21 in the same report — moving walkways can be **suppressed** or **flagged for a single trend**.

Part B — Trends tab (calendar presets)

The Trends tab insight answers: **“What changed in this chart period?”** The user picks one of four presets (`last_week` , `last_2_weeks` , `last_month` , `last_2_months`). The gate flags metrics within that window; **two or more** flagged metrics → LLM; **one** → rich attention template; **zero** → healthy baseline portrait (peer standing, period deltas, sub-bucket stability).



Trends coverage matrix

#	Unit	Type	Building	Preset	UI state	Path	LLM?
B1	PE 08	Hydraulic	EWR Terminal C	<code>last_month</code>	● Healthy	Healthy template	No
B2	PE 08	Hydraulic	EWR Terminal C	<code>last_week</code>	● Attention	Single-metric template	No
B3	PE 08	Hydraulic	EWR Terminal C	<code>last_2_weeks</code>	● Attention	Rich attention template	No

B4	EL-111	Hydraulic	JFKIAT Terminal 4	last_2_months	 Degraded	Trends LLM	Yes
B5	EL-111	Hydraulic	JFKIAT Terminal 4	last_2_weeks	 Attention	Trends LLM	Yes

Key finding: The **same unit (PE 08)** tells different stories by preset — single-metric attention on short windows, healthy portrait on last_month , while **EL-111** escalates to LLM when relevel activity and door metrics compound.

B1. PE 08 — Hydraulic, EWR Terminal C (last_month)

Device ID: 53842FFE-B017-42F3-9BFC-619BC2E6BB5D

The scenario

Period aggregate healthy: relevels down ~60%, door cycle ratio flat at 1.99. L1 shows only 1 incident in the month (spread). Gate: zero flagged metrics → healthy template.

Without the insight card

Manager selects **Last month** on Trends, then scans up to 12 charts for relevels, door ratio, and motor starts to confirm “all quiet.”

What the card showed

Equipment Insight	 Healthy
PE 08 showing stable operation with reduced releve activity over two weeks	
• Relevels declined 60.5% over the past two weeks (from 38 to 15 events). • Door cycle trip ratio holding steady at 1.99 with minimal 0.003% change. • Relevel motor start ratio decreased 54% over two weeks (from 0.0069 to 0.0032).	
Updated 8 Jul, 16:48	

How the insight helps

Healthy Trends cards are **no longer one-liners** — they cite period numbers so the manager can trust “green” without opening every chart.

B2. PE 08 — **last_week** (short window, single trend)

The scenario

Door cycles flagged as falling for 3 weeks within the week preset. One signal → single-metric template path.

Without the insight card

Manager switches preset to **Last week**, finds the door-cycles chart, estimates trend length manually.

What the card showed



How the insight helps

Short-window **nudge** — same role as Overview single-metric templates (cf. A06). Preset change alone shifts which metric leads.

B3. PE 08 — **last_2_weeks** (rich attention + sub-buckets)

The scenario

Average door open time flagged (2-week trend). Sub-bucket logic surfaces **2-day** and **weekly** granularities (Option B). One metric → attention template with multiple bullets + watch item.

Without the insight card

Manager compares door-open chart at daily vs. weekly rollups; easy to miss that 2-day buckets rise while weekly aggregate looks flatter.

What the card showed

Equipment Insight

● Attention

Average door open time shows a sustained rising trend this period.

- Average door open time has been rising for 2 weeks (current 13.2 vs baseline 13.5).
- ... rising pattern across 2-day intervals (2 buckets).
- ... falling pattern across weekly intervals (1 bucket).

Watch: Watch whether average door open time continues rising over the next period.

Updated 8 Jul, 17:01

How the insight helps

Demonstrates **preset-sensitive sub-bucket copy** — richer than B2 without LLM cost.
Manager sees *which granularity* drove the flag.

B4. EL-111 — last_2_months (LLM + degraded)

Device ID: 5A993DD3-9278-43EB-83AD-95567B726355

The scenario

Door cycle time +7% over 3 weeks; **39 unplanned incidents** and 24.3 h downtime in 2 months (L1: `door_locks` clustered). Multiple metrics + incident crossover → Trends LLM, degraded.

Without the insight card

Manager toggles **Last two months**, cross-checks trend charts and incident export separately to link door timing with lock faults.

What the card showed

Equipment Insight⬢ Degraded

Door cycle time rising 7% over 3 weeks; 39 unplanned incidents in 2 months.

- Average door cycle time increased 7.0% to 16.6s ... 39 unplanned incidents ... 24.3 hours downtime ... 13x spike from 3 incidents in the prior period.
- Door lock faults: 21 of 39 incidents, longest cumulative downtime.
- Door cycle trip ratio declined 0.46% to 1.11 ...

Watch: Door lock mechanism—21 incidents in 2 months ...

Watch: Cycle time trend—7% increase over 3 weeks ...

Updated 8 Jul, 17:03

How the insight helps

Strongest **Trends + incident log bridge** in the study — one card connects chart period to outage pattern (door locks), which lives on a different tab.

B5. EL-111 — last_2_weeks (LLM attention)

The scenario

Relevels +55%, relevel motor ratio +47% in two weeks; door metrics mixed; 4 unplanned incidents in the last week vs. 1 prior week.

Without the insight card

Spot relevel spike on one chart; miss concurrent motor-start ratio and incident cluster without tab-hopping.

What the card showed

Equipment Insight

● Attention

Relevel activity and motor starts rising sharply over two weeks.

- Relevels increased 55.4% (590 to 917) ...
- Door cycle trip ratio fell 0.6% while avg door cycle time rose 4.3% ...
- Four unplanned incidents in the last week ...

Watch: Relevel frequency spike ...

Watch: Door lock incidents clustering in current week

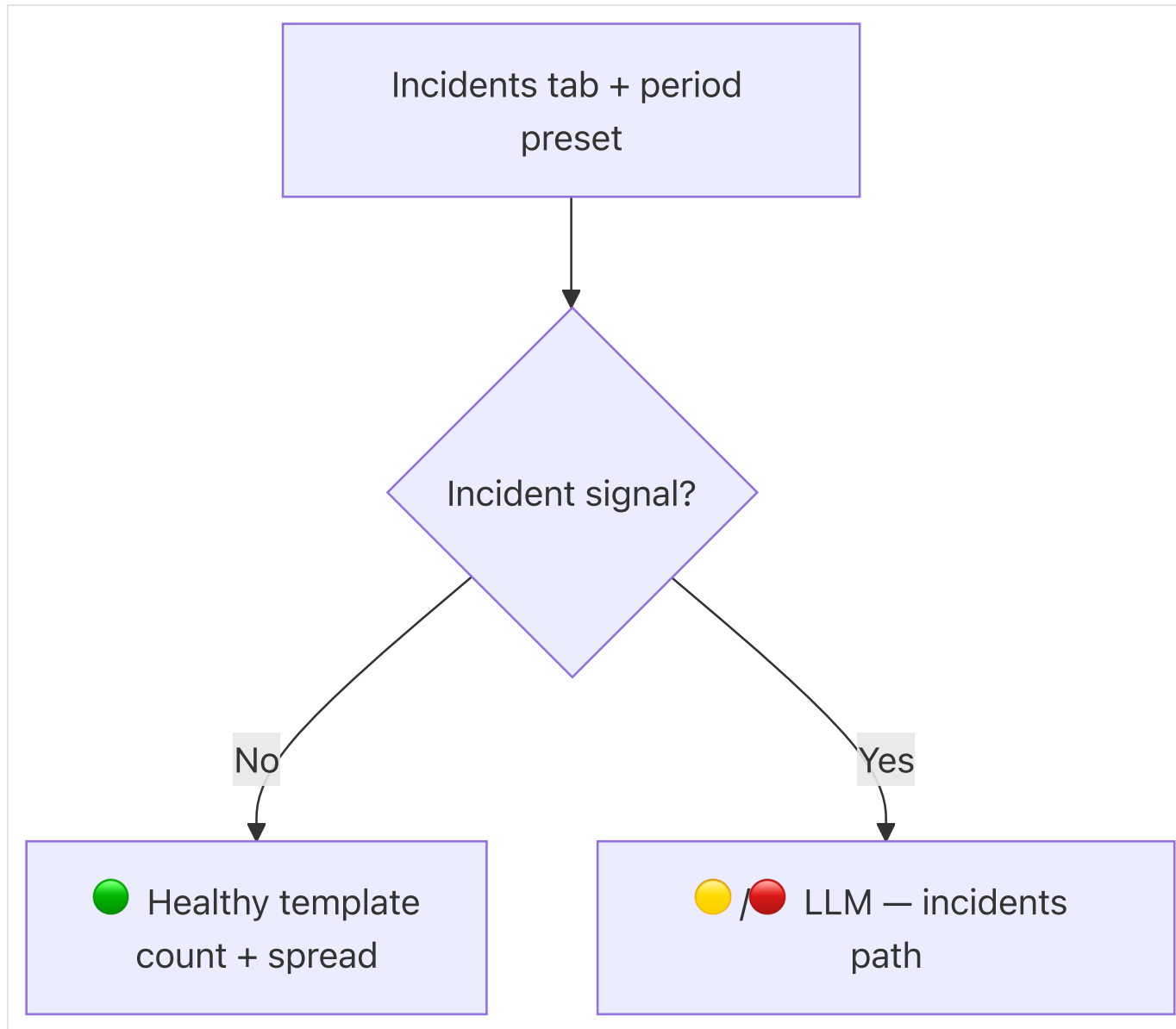
Updated 8 Jul, 17:03

How the insight helps

Same device as B4, **different preset** — short window emphasizes acute relevel surge vs. two-month door-lock story.

Part C — Incidents tab (calendar presets)

The Incidents tab insight answers: **“What pattern shows up in the outage log this period?”**
Gate fires on elevated incident z-score **or** clustered incidents (≥ 2). Any signal → LLM; quiet period → healthy template citing count, downtime, spread.



Incidents coverage matrix

#	Unit	Type	Building	Preset	UI state	Path	LLM?
C1	B03N01 ELEV	Hydraulic	Union Station	last_2_weeks	 Healthy	Healthy template	No
C2	PE 2	Hydraulic	JFK AirTrain Howard Beach	last_month	 Degraded	Incidents LLM	Yes
C3	PE 3	Traction	JFKIA Blue Parking Garage	last_month	 Degraded	Incidents LLM	Yes
C4	EL-111	Hydraulic	JFKIAT Terminal 4	last_month	 Healthy	Healthy template	No
C5	Passenger Elevator #1	Traction	Louisiana Station	last_month	 Degraded	Incidents LLM	Yes

C1. B03N01 ELEV — Union Station (last_2_weeks , healthy)

Device ID: 9FFB8913-49ED-4808-B00A-E39815111840

The scenario

3 incidents, 0.4 h downtime, SAFETY CIRCUIT top fault, **spread** clustering. Below gate threshold for LLM on this preset → healthy template.

Without the insight card

Scan incident table, count rows, estimate downtime manually.

What the card showed

Equipment Insight

● Healthy

Incident activity this period is within the normal range.

- 3 incidents this period, within typical levels.
- Total downtime was 0.4 hours.
- Incidents were spread across the period rather than clustered.

Updated 8 Jul, 17:06

How the insight helps

Negative confirmation on Incidents — manager knows the log is quiet without reading every row.

C2. PE 2 — Howard Beach (last_month , clustered volume)

Device ID: A08226C4-1721-4C2B-227E-515F34E70526

The scenario

L1: **31 incidents**, 9.16 h downtime, gate_switch dominant, **clustered**. +24% vs. prior month (25 → 31). Gate → Incidents LLM.

Without the insight card

Export June incidents, pivot by fault type, check dates for clustering, compare to May count.

What the card showed

Equipment Insight

Degraded

PE 2 experienced 31 unplanned incidents in June, up from 25 prior month.

- All 31 unplanned; gate_switch caused 11 incidents and longest cumulative downtime.
- Frequency up 24% month-over-month, clustered in time.
- Total downtime 9.16 hours, concentrated in unplanned outages.

Watch: Gate switch fault recurrence and root cause
Watch: Temporal clustering – shifts or days

Updated 8 Jul, 17:07

How the insight helps

Turns an **incident table** into a fault-level summary with MoM context — the core Incidents tab value prop.

C3. PE 3 — Blue Parking Garage (last_month , downtime-heavy)

Device ID: E76DE40C-E3FB-45D6-AB25-05D43C1FFA22

The scenario

9 incidents but **55.85 h downtime** (~6.2 h/incident); `door_locks` in 6 of 9; clustered. Spike from 1 incident prior month.

Without the insight card

Mean time to repair hidden in per-row durations; manager might see “only 9” and under-prioritize.

What the card showed

Equipment Insight

Degraded

PE 3 experienced 9 unplanned incidents in June, concentrated in door locks.

- 9 incidents over 55.85 hours ... vs. 1 prior period.
- Door locks: highest frequency (6) and longest cumulative downtime.

Watch: Door lock system recurrence and clustering

Watch: Unplanned incident rate increase from baseline

Updated 8 Jul, 17:09

How the insight helps

Contrasts with C2: **count vs. severity** — LLM highlights downtime concentration, not just incident count.

C4. EL-111 — last_month (healthy spread)

The scenario

7 incidents, 2.6 h downtime, **spread** (not clustered) — gate stays healthy. Same device as B4/B5 but Incidents tab says “within normal range” for this month.

What the card showed

Equipment Insight● Healthy

Incident activity this period is within the normal range.

- 7 incidents this period ...
- Total downtime was 2.6 hours.
- Incidents were spread across the period rather than clustered.

Updated 8 Jul, 17:10

How the insight helps

Cross-tab nuance: EL-111 Trends can be degraded (B4) while Incidents last_month is healthy (C4) — different questions, different gates.

C5. Passenger Elevator #1 — Louisiana Station (last_month)

Device ID: B19F080D-BD3E-4836-AE03-CD1E86E221F8

The scenario

25 unplanned incidents, **285.76 h downtime**, all `door_zone` faults, clustered; up from 21 prior month.

What the card showed

Equipment Insight

Degraded

Elevator #1 experienced 25 unplanned incidents in June with concentrated door zone faults.

- All 25 unplanned; 285.76 hours downtime.
- Door zone faults: all 25 incidents, top frequency and duration.
- Up from 21 prior month; clustered in time.

Watch: Door zone mechanism—all faults cluster here

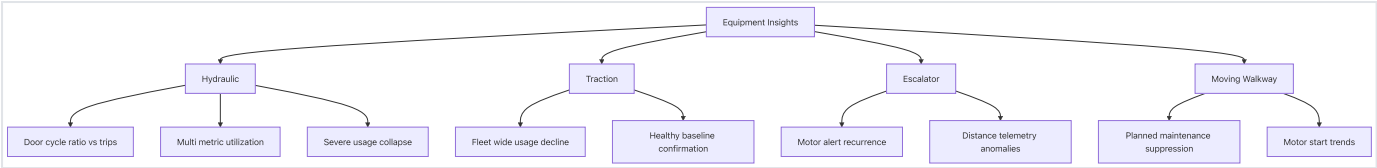
Watch: Downtime trend—285+ hours in one month

Updated 8 Jul, 17:11

How the insight helps

Extreme **single-fault dominance** — manager immediately knows the outage story is door-zone, not a mix of unrelated codes.

Insights by equipment type



Equipment type	Dominant signals in this study	Best insight value
Hydraulic	Door cycles, trip ratio, cycle time, runtime	LLM when door behavior diverges from trip volume (A01); Trends sub-buckets on door open time (B3); door-lock incident chains (B4, C3)
Traction	Trips, distance, motor starts, runtime	LLM when many metrics move together (PE-A #2, Elevator D); high-downtime incident digest (C3, C5)
Escalator	Motor alerts, distance, runtime	LLM linking alerts + telemetry (ES-HB NO. 1- UP)
Moving walkway	Suppression state, motor starts	Template paths — simple and appropriate

When LLM vs. template — decision summary

Overview (Part A)

Customer situation	Path	Example from study
"Nothing to see"	Template	Elevator #15
"One thing shifted"	Template	A06, CB-MW02
"We're maintaining it"	Template	BW-21
"Several things changed together"	LLM	A03, A04, PE-A #2, A01
"Alerts + metrics tell one story"	LLM	ES-HB NO. 1- UP
"Something serious / contradictory"	LLM + Degraded	Elevator D

Trends (Part B)

Customer situation	Path	Example
All metrics in range for preset	Healthy template	B1 PE 08 last_month
One metric trend in window	Attention template	B2, B3 PE 08
Two+ metrics or ratio + volume story	LLM	B4, B5 EL-111

Incidents (Part C)

Customer situation	Path	Example
Low volume, spread	Healthy template	C1, C4
Elevated count or clustering	LLM	C2, C3, C5

Rule of thumb for customers: If the card has **watch items** and **multiple numbered signals with baselines**, the LLM did the cross-chart or cross-row work for you. Rich **healthy** and **attention templates** now also show multiple bullets — check [outcome](#) in the API or whether watch items appear without a multi-paragraph headline.

Limitations and gaps (this study)

Gap	Notes
Active safety headline	Elevator D showed metric collapse, not a safety-specific card. A dedicated safety scenario remains useful for a future pass.
Snoozed (non-planned)	Only planned maintenance suppression was captured (BW-21).
Alert-only single signal	No device in Overview set had exactly one alert flag with zero metric flags.
Live vs. cached drift	Several units changed state between 1 Jul (DB) and 6–8 Jul (UI). Insights are live — expected behavior.
Trends preset sensitivity	PE 08 healthy on <code>last_month</code> but attention on <code>last_week</code> — same device, different cards (B1 vs B2).
Cross-tab divergence	EL-111 degraded on Trends <code>last_2_months</code> (B4) but healthy on Incidents <code>last_month</code> (C4) — tabs answer different questions.
Incidents cache cold start	First Incidents capture required API/UI hit per device after cache clear; L1 (<code>device_period_stats</code>) was pre-populated by cron.
Escalator / MW on new tabs	Part B/C devices are hydraulic/traction only; extend with ES-HB and CB-MW02 in a follow-up pass.

Methodology

Part A — Overview (6 Jul 2026)

- Device selection:** Queried Jobs DB `device_insight_cache` and snapshot tables for diverse outcomes across equipment types (see `sql-queries_llm.txt` , queries q1–q9).
- Metadata enrichment:** Joined device IDs to `Twins_Backup` for nickname, building, and equipment type.
- UI capture:** Opened each device on the staging dashboard and copied the full insight card text (see `llm-insights.txt` header section / Overview captures).

- 4. **Analysis:** Compared underlying flagged signals (DB) to rendered card text (UI) to assess customer value.

Part B & C — Trends and Incidents (8 Jul 2026)

- 1. **Prerequisites:** Migration `005_equipment_insights_v1.sql` on jobs DB; cron populated `device_period_stats` for all four presets; API deployed with tab-aware cache `(device_id, tab, period_preset)`.
- 2. **Device selection:** Queried `device_insight_cache` and `device_period_stats` by tab (see `sql-queries_llm.txt`, section “new query results” q1–q7). Picked devices maximizing outcome diversity (healthy, single template, rich attention, LLM degraded).
- 3. **Cache reset:** `DELETE FROM device_insight_cache WHERE tab IN ('trends', 'incidents')` before capture to force fresh templates/LLM after code fixes.
- 4. **UI capture:** Opened Trends or Incidents tab, selected calendar preset, copied card text into `llm-insights.txt` (8 Jul timestamps).
- 5. **API:** `GET /api/equipment/{id}/insight?tab=trends|incidents&period_preset=last_week|last_2_weeks|last_month|last_2_months`

Appendix — device reference

Part A — Overview (6 Jul 2026)

Device ID	Nickname	Type	Building
<code>EC4F9F1B-...</code>	A06	Hydraulic	EWR Terminal A
<code>E292092B-...</code>	Elevator #15	Traction	Woolworth Building

475A4B8B-...	A03	Hydraulic	EWR Terminal A
9191D75D-...	A04	Hydraulic	EWR Terminal A
2509AA46-...	PE-A #2	Traction	JFKIA PA Bldg. 14
3ADEB45E-...	A01	Hydraulic	EWR Terminal A
4EE9BD2E-...	ES - HB NO. 1- UP	Escalator	JFK AirTrain Howard Beach
5B9A752F-...	BW-21	Moving walkway	Concourse B
1CECA405-...	Elevator D	Hydraulic	World Trade Center
908F0F62-...	CB-MW02	Moving walkway	LGA Terminal B

Part B & C — Trends / Incidents (8 Jul 2026)

Device ID	Nickname	Type	Building
53842FFE-B017-42F3-9BFC-619BC2E6BB5D	PE 08	Hydraulic	EWR Terminal C — United
5A993DD3-9278-43EB-83AD-95567B726355	EL-111	Hydraulic	JFKIAT Terminal 4
A08226C4-1721-4C2B-227E-515F34E70526	PE 2	Hydraulic	JFK AirTrain Howard Beach
9FFB8913-49ED-4808-B00A-E39815111840	B03N01 ELEV	Hydraulic	Union Station
E76DE40C-E3FB-45D6-AB25-05D43C1FFA22	PE 3	Traction	JFKIA Blue Parking Garage
B19F080D-BD3E-4836-AE03-CD1E86E221F8	Passenger Elevator #1	Traction	Louisiana Station

Related documentation: equipment_insights_brief.md ,
equipment_insights_frontend_llm_eval.md ,
equipment_insights_data_findings.md